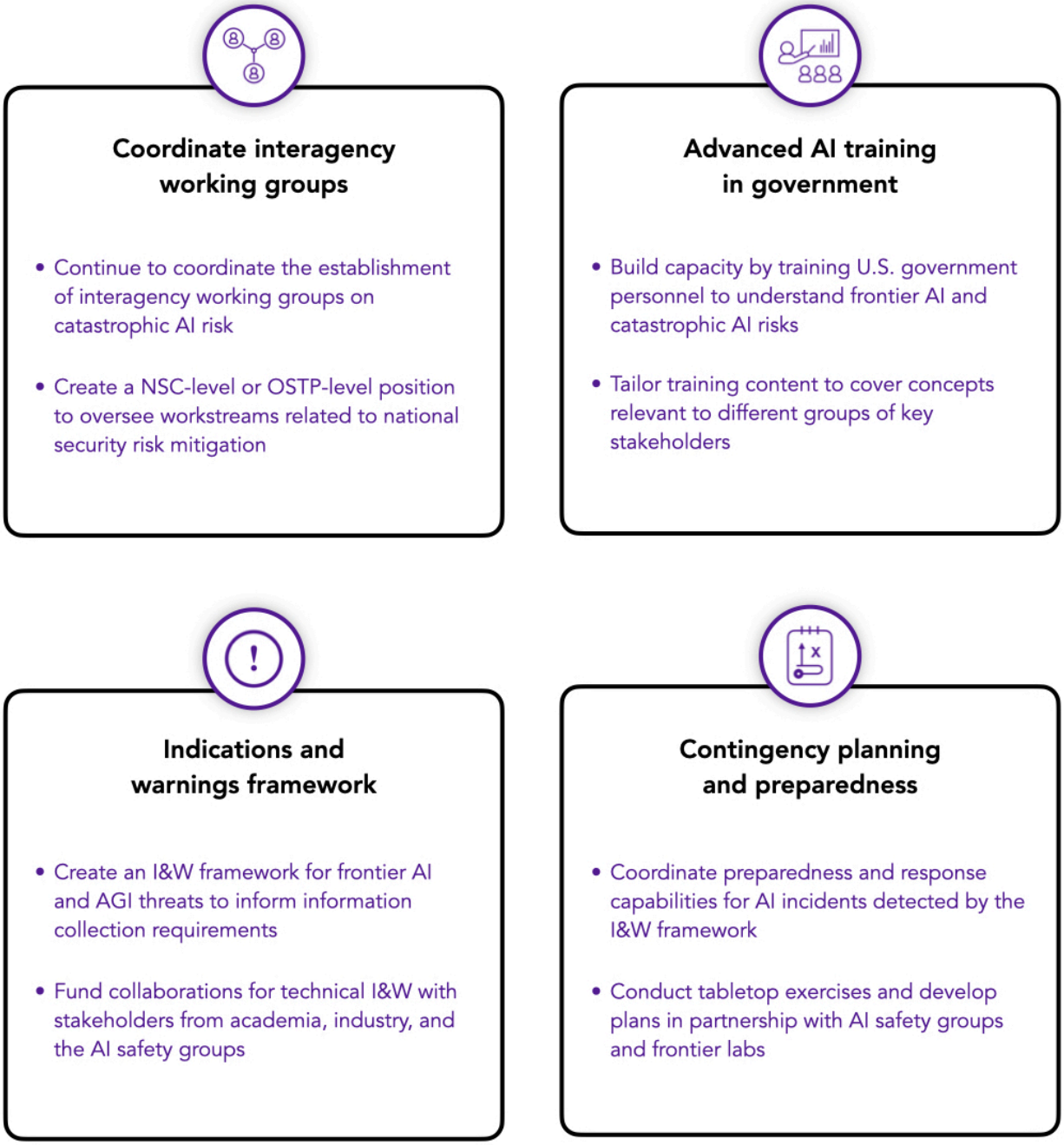
LOE2: Strengthen capability and capacity for advanced AI preparedness and response



Advanced AI and AGI risk mitigation will engage a broad set of U.S. government equities. However, the advanced AI landscape is changing rapidly (Introduction, 0.1). In this climate, policy should support swift, well-coordinated decision-making grounded in rigorous and timely technical assessments (Introduction, 0.5.2.1). Effective mitigation

measures will require advance planning and accurate assessments and synthesis of risk signals in a potentially challenging information environment (Introduction, 0.5.2.2). Execution in these areas, and in those of the other LOEs, will necessitate substantial institutional capacity-building across all U.S. government stakeholders.

This LOE outlines specific actions that the U.S. government could take to **increase its preparedness for rapidly responding to incidents related to advanced AI and AGI development and deployment**. These actions are:

- Continuing to establish and coordinate **interagency working groups**, including for the LOEs in this action plan (see 2.1);
- Increasing preparedness and response capacity and capability through **education and training** (see 2.2);
- Coordinating the development of an **Indications and Warnings (I&W)** framework for advanced AI and AGI incidents (see 2.3); and
- Coordinating the development of scenario-based **contingency plans** (see 2.4).

The I&W and contingency planning processes will generate advanced AI-related risk management requirements that could inform other aspects of this plan, including the specific thresholds for RADA safeguards (LOE1, 1.3.2; LOE4, 4.1.3).

2.1 Coordinate interagency working groups

We recommend the U.S. government continue to coordinate the establishment of interagency working groups to execute on mitigation measures for catastrophic AI risk. To ensure such measures are treated as a holistic and coherent effort, the President could create an NSC-level or OSTP-level position with responsibility for all workstreams related to national security risk mitigation from advanced AI and AGI. This could include overseeing the execution of some of the LOEs in this action plan through the participating departments and agencies. For example the NSC-level official could:

- Receive regular reports on the state of the advanced AI landscape, including capabilities and risk assessments produced by an AIO (DHS; LOE1, 1.2);
- Oversee an ASTF with the mission to implement RADA safeguards for domestic frontier AI (DOE or DHS; LOE1, 1.4);

- Oversee the implementation of I&W (U.S. Intelligence Community; see 2.3) and contingency planning efforts (see 2.4) designed to guard against potential catastrophic risks from advanced AI and AGI;

- Support the coordination of regulatory efforts (LOE1, 1.4; LOE4, 4.1) with federally funded advanced AI and AGI safety and security research (National Science Foundation; LOE3, 3.1);

- Ensure regulatory efforts are aligned with emerging advanced AI and AGI safety and security standards (NIST U.S. AISI; LOE3, 3.2);

- In the long term, oversee a permanent regulatory agency for advanced AI (LOE4, 4.1); and

- Oversee U.S. efforts to build international capacity, consensus, and controls to manage catastrophic AI risks (Department of State; LOE5).

2.2 Advanced AI education and training

AI safety and security is a complex field in which nuanced technical factors often make the difference between effective and ineffective risk mitigation strategies. U.S. government personnel charged with implementing or developing AI safety policy must understand these factors in order to fulfill their duties successfully. However, different personnel will have varying educational requirements depending on their domain of work and position.

The U.S. government will therefore require a carefully targeted educational program, designed to provide training and support of the right type and depth to individuals charged with advancing U.S. national security interests in AI. This program should have several key properties:

1. **Ensure that the training and support it provides are updated regularly.** This is necessary to account for rapid advances in AI capabilities and risks as the frontier of the field progresses.

2. **Introduce ongoing requirements for continuous learning.** Significant advances that shape the landscape of AI capabilities occur on at least a monthly basis [160]. Without continuous learning, even trained personnel would quickly find themselves with an outdated understanding of frontier AI capabilities and risks.

In many cases this knowledge gap could pose a risk to their ability to execute their missions effectively.

3. **Deliver the right information to the right stakeholders.** Personnel involved in commercial regulation, military safety policy, international diplomacy, and other fields will each require different training focused on the aspects of the frontier AI landscape relevant to each of their respective domains.

4. **Deliver information in the right format for its intended audience.** Most relevant personnel can likely be trained via asynchronous online courses. But others may require live in-person training, ongoing access to centers of expertise within or outside government, or combinations of these options.

A robust, regularly updated, and technically grounded training program will be critical to support government decision-making in areas such as:

- Developing technically-informed policy, regulatory, and legal frameworks;
- Uncovering requirements for necessary new authorities and programs;
- Contributing to technical advances in AI safety and security; and
- Messaging the topic clearly and effectively with the American public.

There are many means by which training can be acquired, including from the private sector, but in all cases it should be vetted for quality and accurate risk focus.

2.2.1 Key stakeholders and learning outcomes

Educational requirements will vary among U.S. government personnel. Below, we provide a list of key stakeholder groups, along with recommended high-priority learning objectives for each. OMB could consider directing agency heads to identify and train personnel in these key areas. We also recommend that the Administration's key influencers acquire their own AI training for the purpose of increasing the technical accuracy of public discourse (Introduction, 0.5.2.2).

It is especially crucial to train technical personnel in defense and national security roles about this risk category, both in the United States and internationally to the fullest extent possible. These individuals need to be equipped to offer their leadership

accurate and timely advice that will inform national security strategy with respect to advanced AI (LOE5, 5.2.1.2).

Table 1. U.S. government stakeholders and potential target learning outcomes of AI education.

Key stakeholders	Target learning outcome
NSC, OSTP, OMB, department and agency political leaders and possible regulators involved in advanced AI policy	Strategic and technically-informed decision-making
AI Observatory personnel (LOE1, 1.2)	Strategic and technically-informed decision-making
AI Safety Task Force personnel (LOE1, 1.3)	Deepened baseline operational understanding prior to negotiating with labs, understand key considerations for enforcement of AI regulations, understand the risks associated with frontier AI research to incentivize responsible behavior
Intelligence Community-led Indications and Warnings workstream and direct management personnel (see 2.3)	Strategic and technically-informed decision-making
Contingency planning workstream and direct management personnel (see 2.4)	Strategic and technically-informed decision-making
Interagency teams working all LOEs	Strategic and technically-informed decision-making
Congressional leaders and staff	Strategic and technically-informed decision-making
Key influencers [161]	Enhance the accuracy of public discourse
Judiciary leaders and staff [162]	Enhance capability to address cases on the range of legal issues presented by advanced AI, frontier AI, and AGI
Department of Justice stakeholders responsible for enforcement of advanced AI regulations (LOE4, 4.1.2.3 and 4.2.2)	Understand key considerations for enforcement of AI regulations
Key diplomats including embassy staff (LOE5, 5.2)	Accelerate U.S. ability to drive a common understanding of potential catastrophic AI risks for international coordination

International partners and stakeholders (LOE5, 5.2.1)	Accelerate U.S. ability to drive a common understanding of potential catastrophic AI risks for international coordination
AI researchers at National Labs and other technical centers of excellence in the U.S. government, and in governments around the world (LOE3, 3.1.2; LOE5, 5.2.1.2)	Understand open problems in technical AI safety and AGI alignment
Private sector AI capabilities researchers	Understand the risks associated with frontier AI research to incentivize responsible behavior
Public sector AI academics (LOE1, 1.5.4 and 1.5.5)	Understand the risks associated with frontier AI research to incentivize responsible behavior

See the next section for examples of training topics that support each of the training outcomes above.

2.2.2 Suggested training topics

Stakeholder training will need to be accessible to non-technical audiences. At the same time, it should be substantive enough to allow personnel to make informed predictions about near-future AI capabilities. It should also equip personnel to interpret semi-technical articles published by frontier labs announcing new breakthroughs (e.g., [163]), and to correctly infer some of their implications.

All stakeholders should obtain a **baseline understanding** of AI and key issues at play in frontier AI safety and security. This baseline understanding should include the following concepts:

- Drivers of AI progress (e.g. AI scaling and algorithmic improvements);
- Bottlenecks to AI progress (e.g. AI-optimized hardware, data, talent);
- Key sources of catastrophic AI risk (including weaponization and loss of control due to AGI alignment failure; Introduction, 0.2.1 and 0.2.2);
- Technical risk mitigation strategies (e.g. AI evaluations, mechanistic interpretability);

- Challenges to risk mitigation (e.g. outcome severity, timelines, incentives, supply chain factors; Introduction, 0.5); and

- Policy-based risk mitigation strategies (e.g. compute-based reporting thresholds, reporting requirements for large-scale training runs).

The training could also inform decisions on policy and safeguards for advanced AI systems; ideas for improving safety standards; and associated economic, antitrust, and regulatory capture concerns. Finally, training could examine operational challenges and solutions for international AI safeguards, aiming to build capacity and capability for implementations of LOE5. Training content could also be packaged as reference documents for government stakeholders. Like the training itself, these documents should be kept frequently updated given the pace of progress in frontier AI.

Below is a list of key training outcomes, along with sample training topics that we recommend as being supportive of each key outcome.

Table 2. Training topics associated with target learning outcomes.

Target outcome of training	Sample training topics
Strategic and technically-informed decision-making	• AI scaling and its relationship to AI capabilities and alignment • AI capabilities and impacts forecasting • Inner vs outer alignment and the stability-control paradox [161] • The open-source AI ecosystem • Compute as a strategic resource • Governance and control of advanced AI systems • The advanced AI supply chain

Deepened baseline operational understanding prior to negotiating with labs	• AI scaling and its relationship to AI capabilities and alignment • Weaponization, accident, and loss of control risks, including catastrophic risks • AI capabilities and impacts forecasting • The open-source AI ecosystem • Compute as a strategic resource • Antitrust and regulatory capture concerns • Concerns over gaming of safety evaluations • The culture of and ongoing debates within the AI alignment community
Enhance the accuracy of public discourse	• AI scaling and its relationship to AI capabilities and alignment • Weaponization, accident, and loss of control risks, including catastrophic risks • AI capabilities and impacts forecasting • Antitrust and regulatory capture concerns • The culture of and ongoing debates within the AI alignment community
Enhance capability to address cases on the range of legal issues presented by advanced AI and AGI	• Weaponization, accident, and loss of control risks, including catastrophic risks • AI capabilities and impacts forecasting • Antitrust and regulatory capture concerns • Open questions in generative AI copyrighting, and how these impact the data collection strategies of frontier labs
Accelerate U.S. ability to drive a common understanding of the risks for international coordination	• AI scaling and its relationship to AI capabilities and alignment • Weaponization, accident, and loss of control risks, including catastrophic risks • AI capabilities and impacts forecasting • Inner vs outer alignment and the stability-control paradox [161] • Concerns over gaming of safety evaluations • Governance and control of advanced AI systems • The advanced AI supply chain

Understand key considerations for enforcement of AI regulations	• AI scaling and its relationship to AI capabilities and alignment • Weaponization, accident, and loss of control risks, including catastrophic risks • AI capabilities and impacts forecasting • Concerns over gaming of safety evaluations • Governance and control of advanced AI systems • The frontier AI ecosystem and its key players • The ideological motivations of frontier AI researchers
Understand open problems in technical AI safety and AGI alignment	• Research agendas aimed at solving AI alignment • Techniques for measuring and mitigating power-seeking behavior • Corrigibility techniques • Mechanistic interpretability techniques • Empirical techniques such as activation engineering, representation engineering, and probing • Expected failure modes of reinforcement learning from human feedback and other prosaic alignment techniques in AGI-level systems
Understand the risks associated with frontier AI research to incentivize responsible behavior	• Weaponization, accident, and loss of control risks, including catastrophic risks • Concerns over gaming of safety evaluations • Research agendas aimed at solving AI alignment • Techniques for measuring and mitigating power-seeking behavior • Corrigibility techniques • Mechanistic interpretability techniques • Empirical techniques such as activation engineering, representation engineering, and probing • Expected failure modes of reinforcement learning from human feedback and other prosaic alignment techniques in AGI-level systems

2.3 Indications and warnings

Without a capacity to detect and anticipate potential catastrophic risks from AI weaponization or loss of control, the U.S. government risks either being caught at a disadvantage by fast-moving national security threats (Introduction, 0.5.1.1, 0.5.1.2, and 0.5.2.1), or causing premature harm to its own domestic AI industry (Introduction, 0.4.2).

Therefore, as a critical near-term action, the U.S. government should develop an indications and warnings (I&W) framework for advanced AI and AGI threats.[42] Development of this framework could be undertaken by the U.S. Intelligence Community (IC).

Once developed, an I&W framework could inform downstream requirements to comprehensively monitor advanced AI research and development globally (LOE1, 1.2), track the release of open-access AI models (Annex D, D.3), map the global supply chains for AI compute hardware and data centers (Annex G, G.3), and assess potential risks from both declared and undeclared frontier AI labs and other projects. The I&W framework should be kept regularly updated as new information becomes available, including by seeking input from frontier labs and the broader AI safety community.

An I&W framework could enable the U.S. government to receive timely warning of emerging risks from frontier AI development programs. Partners such as DHS and DOE could also help increase the government's understanding of technical I&W for advanced AI and AGI threats through collaborations with domestic frontier labs. The I&W framework could also benefit from coordination and information-sharing with other task forces and agencies such as the AIO (LOE1, 1.2), ASTF (LOE1, 1.4), federally funded AI research centers (LOE3, 3.1.2) and regulatory bodies (LOE4, 4.1). Such partnerships could support the IC in ensuring that the framework remains informed by an up-to-date and accurate view of the frontier AI landscape.

Traditional approaches to I&W [164] may need to be modified to accommodate challenges associated with modeling catastrophic AI risk. These could include the practical difficulties of accessing qualified technical talent to advise on I&W

[42] Advanced AI threats are not necessarily AGI threats. AI systems could pose catastrophic weaponization risk before AGI is developed, for example.

development, and the technical difficulties of forecasting the emerging capabilities of future frontier AI systems.[43]

2.3.1 Key categories and sources of catastrophic risk

Below is a preliminary taxonomy of entities and activities that, either now or in the medium-term future (1-5 years as of December 2023), could introduce catastrophic AI risks. See Introduction, 0.3 for descriptions of these entity categories and Annex D: Advanced AI landscape for more details. See Introduction, 0.2.1 and 0.2.2 for more details on the weaponization and loss of control risk categories.

Table 3. Potential sources of catastrophic AI risk, corresponding risk categories, and information requirements associated with each.

Potential risk sources	Potential risk categories
Domestic frontier AI programs	• Accidental emergence of dangerous capabilities or competent power-seeking behavior (**loss of control**) [1] • Sudden capability jumps enabled by external software frameworks (**loss of control**) [120] • Exfiltration followed by weaponization of stolen model by an adversary (**weaponization**) • Undetected misuse by end-users (**weaponization**)

[43] Although the emergence of specific AI capabilities cannot yet be reliably predicted in advance [165], it may still be possible to develop imperfect early indicators. For example, METR [166] is exploring an approach of breaking a dangerous capability down into a set of sub-tasks, and then measuring AI model performance on those sub-tasks. For example, in attempting to anticipate a model's capability to execute autonomous cyberattacks, that capability could be broken down into sub-tasks like the ability to write short code snippets or entire malware files, the ability to autonomously probe a cyber environment for vulnerabilities, and so on. See LOE3, 3.1 for recommendations on supporting research in this area.

Foreign AI programs	• Accidental emergence of dangerous capabilities or competent power-seeking behavior (**loss of control**) • Sudden capability jumps enabled by external software frameworks (**loss of control**) • Undetected misuse by end-users (**weaponization**) • Intentional offensive deployment (**weaponization**)
Open-access release of advanced AI models	• Accidental emergence of dangerous capabilities via fine-tuning (**loss of control**) • Sudden capability jumps enabled by external software frameworks (**loss of control**) • Intentional weaponization via fine-tuning or direct use (**weaponization**)
Theft or sale and subsequent augmentation of frontier AI models by state or non-state actors	• Accidental emergence of dangerous capabilities or competent power-seeking behavior after fine-tuning (**loss of control**) • Sudden capability jumps enabled by external software frameworks (**loss of control**) • Intentional offensive deployment (**weaponization**)

An I&W framework will need to consider a broad range of scenarios linked to these entity categories, and could be informed by many different sources of data [167–175]. The I&W framework should consider scenarios in which risk primarily emerges from an AI model or AI system itself, such as loss of control due to AGI alignment failure [176]. They should also include scenarios in which risk derives from specific actions by end-users or model developers, such as automated large-scale cyberattacks. The resulting I&W framework should in turn inform contingency planning efforts (see 2.4). Example scenarios could include:

- The unexpected emergence of dangerous AI capabilities at a U.S. or Western frontier lab [177];

- An open-access AI system fine-tuned to support cyber or CBRN attacks;

- The sudden discovery of a previously unrecognized capability in a generally available open-access AI model that enables dangerous weaponized applications;

- The uncontrolled deployment of an AGI or near-AGI system by a hedge fund that autonomously engages in profit-driven information warfare [178];[44]

- Systemic vulnerabilities introduced by integrating interdependent AI systems into decision-making processes whose interactions may have unpredictable effects (e.g. a flash crash in the financial system);

- Convincing evidence that suggests that AGI alignment is an intractable problem, or that it cannot be expected to be solved before AGI is developed [7]; or

- A breakthrough that makes it possible for a particular weaponized application to be deployed below a certain budget.

For each scenario, the I&W framework could maintain and frequently update a set of key metrics, for example:

- The estimated breakout timelines to various levels of AI capability by identified domestic and foreign entities;

- The largest AI training run a domestic or foreign entity could undertake in a set amount of time; and

- The likelihood of emergence of high-risk capabilities under various AI training and deployment regimes.

The I&W framework could also consider longer-timescale metrics that do not need to be updated as frequently. For example, understanding how long it may take a country to develop an independent AI supply chain under various assumptions, and which critical inputs need to be controlled to prevent that development [172].

As part of the I&W framework development process, the U.S. government should consider convening an interagency workshop aimed at mapping the AI ecosystem and AI hardware supply chain, its key domestic and international players, and the activities

[44] See Annex D, D.4 for more information about this risk category.

in which they are engaged that most directly contribute to potential catastrophic risks from weaponization and loss of control.

2.3.2 Improving I&W through bug bounties

Software frameworks such as Auto-GPT [120] and BabyAGI [122] extend the capabilities of existing AI systems in unpredictable ways by helping them perform complicated multi-step tasks without any additional training. They therefore introduce new risks that AI developers are not equipped to anticipate or assess. Some of these frameworks, like the infamous ChaosGPT [121], are intended to deliberately steer the underlying AI system toward destructive behaviors.

To ensure an I&W framework remains robust over time, it is crucial to improve understanding of how software frameworks like these can impact downstream AI capabilities. These kinds of impacts can be assessed openly, so the government can offer bug bounties and other incentives to support this effort. These incentives could encourage individuals and teams to report issues, vulnerabilities, and unsafe behaviors in frontier AI systems. They could also serve to augment AI developers' existing bug bounty programs.

2.3.3 Improving I&W through direct government investment

Currently, no one knows how to forecast the emergence of high-risk AI capabilities under different training and deployment conditions. This lack of knowledge increases the uncertainty and challenge of developing and maintaining a robust I&W framework.

To close this gap, the IC should consider establishing a formal academic consortium or cooperative research agreement for AI safety and security and technical I&W. This consortium could leverage top universities to help refine forecasting approaches by advancing the state of the art in areas such as AI capability evaluations, transparency and interpretability, and capability prediction techniques (LOE3, 3.2).

In addition to an academic consortium, the U.S. government could establish an explicit vehicle for collaboration between the IC, frontier labs, and private sector AI safety and AGI alignment research organizations to inform the I&W development and maintenance process. The commercial market for this research remains limited, and

government investment could accelerate efforts to improve the technical accuracy of AI forecasting in support of robust I&W.[45]

Either or both of the above initiatives could be coordinated through federally funded AI research centers dedicated to this problem set (LOE3, 3.1.2.3). The IC could also consider leveraging the AI testbeds and infrastructure under development by DOE and NIST to support I&W framework development and implementation. These testbeds could be used to run evaluations on AI models that map directly onto key information requirements associated with the I&W framework (see 2.3.1). This could include developing model evaluations specifically tailored to the I&W use case.

2.4 Contingency planning

Under some worst-case scenarios (see 2.3.1), decision-makers could have only a very short window (hours to weeks) to react effectively to a catastrophic threat. An I&W framework (see 2.3) could support the detection of early warning signs for such scenarios, but contingency planning will be essential to allow for the development of appropriate responses.

While no plan can anticipate all contingencies, the planning process itself is essential to developing a common understanding of crisis scenarios and response options. These options can help the government surface detailed requirements for new authorities, programs, resources, policies, and coordination mechanisms. Therefore another critical and immediate task for the U.S. government should be to direct, through the NSC, an interagency contingency planning process to develop response options for various scenarios and timelines of frontier AI development.

Planning could be informed by a range of scenario-based tabletop exercises (TTXs) covering advanced AI risks and sources, up to and including how a safe and secure AGI should be governed if and when it is developed. It may be necessary to collaborate in these exercises with stakeholders from frontier labs, AI safety groups, and evaluation and red teams to ensure the parameters of the exercises at all times remain as realistic as possible [161]. It may also be helpful to run joint exercises such as crisis response or lab shut-down drills with data center infrastructure providers, AI hardware owners, and frontier labs to anticipate and better understand key contingencies.

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[45] Another reason to favor U.S. government funding for these organizations is that the great majority of them are nonprofits, and are currently funded by the same set of large donors. See Annex E: Funding in AI safety for more information.

Contingency planning activities should involve the full range of stakeholders within the U.S. government who would be involved in implementing emergency responses to critical events and risks. They could account for input from an AIO (LOE1, 1.2), ASTF (LOE1, 1.4), federally funded AI research centers and relevant collaborations, (LOE3, 3.1.2.2), and eventual regulatory bodies (LOE4, 4.1). In determining which events to plan for, planners should consider the set of scenarios uncovered in the I&W development process (see 2.3), and identify any additional scenarios that they consider necessary to address.[46]

Planners should flag these additional scenarios to stakeholders in the I&W development process when possible. We recommend that contingency planning proceed iteratively, to allow initial planning activities to surface information requirements which, when met, would allow future planning and I&W development activities to be more technically informed.

2.4.1 Updating the National Preparedness System

In the event that a severe incident occurs as a result of advanced AI weaponization or loss of control in highly capable AI systems, early planning for prevention and incident response will be critical.[47] We therefore recommend that DHS direct FEMA to update its National Preparedness System [183] to address advanced AI and AGI risk. These updates could be executed in collaboration with an AIO (LOE1, 1.2.1), ASTF (LOE1, 1.4), or eventual regulatory body (LOE4, 4.1).

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[46] For example, see the companion document, **Survey of AI Technologies and R&D Trajectories**, for three sample AGI risk scenarios.

[47] It is possible that genuine AGI loss-of-control scenarios would not be survivable events (Introduction, 0.5.1.1), which if true would remove the need for incident response. But this is currently uncertain and should not be assumed in advance.